MIL-C-44103A
6 August 1985
SUPERSEDING
MIL-C-44103
27 June 1983

MILITARY SPECIFICATION

CLOTH, DUCK, POLYESTER, FIRE, WATER AND WEATHER RESISTANT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers polyester duck treated for resistance to fire, water, and weather.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-1135 - Packaging of Duck Fabrics (Cotton; Synthetic Fiber; Cotton-Synthetic Fiber Blends)

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

OTHER GOVERNMENT DOCUMENTS

Laws and Regulations:

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1424 - Tear Resistance of Woven Fabrics by the Falling-Pendulum (Elmendorf) Apparatus
D 523 - Specular Gloss

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Copies may be obtained from the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

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3. REQUIREMENTS

3.1 Standard sample. The dyed and finished cloth shall match the standard sample for shade and finish appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 Fiber. The fiber shall be regular tenacity, semi-dull fiber prepared from polyethylene glycol terephthalate with a minimum melting point of 472°F.

3.3 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.2.3.

TABLE I. Physical requirements

Characteristic	Requirements	
	Minimum	Maximum
Weight, oz/yd^2:	---	13.5
Yarns per inch:		
Warp	44	---
Filling	32	---
Breaking strength, pounds:		
Warp	400	---
Filling	300	---
Tearing strength, pounds:		
Warp	11	---
Filling	7	---
Air permeability, ft^3/min/ft^2:	---	1.0
Gloss (face side only), percent:		
60 degree specular gloss	---	2.0
85 degree specular gloss	---	2.0

3.3.1 Weave. The weave shall be plain. The cloth shall be woven on either a fly-shuttle or a shuttleless loom. When the cloth is woven on shuttleless looms, the fringed selvages shall be removed from the cloth.

3.4 Width. The width of the cloth shall be as specified (see 6.2) and shall be the minimum width inclusive of the selvage when fly shuttle looms or shuttleless with tuck-in selvage looms are used. For all other shuttleless

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looms the width measurement shall be made between the last warp yarn on each side excluding the protruding fringe(s).

3.5 Color. The color of the finished cloth shall be as specified (see 6.2) and shall match the standard sample.

3.5.1 Matching. The color and appearance of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps which approximate artificial daylight having a correlated color temperature of 7000 ± 500K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 100K.

3.5.1.1 Spectral reflectance. When Forest Green 433 is specified, the spectral reflectance factors (in percent) for visible/near infrared wavelength range of 600 to 860 nanometers of the finished cloth shall meet the requirements specified in table II. When Desert Tan 459 is specified, the spectral reflectance factors (in percent) for the visible/near infrared wavelength range of 700 to 860 nanometers of the finished cloth shall meet the requirements specified in table III.

TABLE II. Spectral reflectance (percent) limits for the Forest Green 433

Wavelengths Nanometers	Reflectance Max	Min	Wavelengths Nanometers	Reflectance Max	Min
600	14	3	740	52	32
620	14	3	760	62	42
640	14	3	780	70	50
660	15	4	800	74	54
680	17	6	820	76	56
700	27	10	840	77	57
720	40	20	860	78	58

TABLE III. Spectral reflectance (percent) limits for Desert Tan 459

Wavelengths Nanometers	Reflectance Max	Min
700	65	52
720	65	55
740	65	55
760	65	55
780	65	55
800	65	55
820	65	55
840	65	55
860	65	55

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3.5.2 Colorfastness. The finished cloth shall show fastness to accelerated weathering equal to or better than the standard sample when tested as specified in 4.2.3. When a standard sample is not available or is not referenced as a standard for colorfastness, the dyed and finished cloth shall show "good" fastness to accelerated weathering when tested as specified in 4.2.3.

3.5.2.1 Crocking. The finished cloth shall show no more crocking on dry white cloth than an AATCC Chromatic Transferance Scale rating of 1.5 when tested as specified in 4.2.3.

3.6 Finish. The cloth shall be given an approved finish (see 6.4, and 6.5).

3.6.1 Water repellent finish. The cloth shall be water repellent treated. The finished cloth shall conform to the requirements shown in table IV when tested as specified in 4.2.3.

TABLE IV. Water repellency requirements

Characteristic	Requirement	
	Minimum	Maximum
Spray rating:		
Initial	90, 90, 90	--
After weathering	70, 70, 70	--
Hydrostatic resistance in centimeters:		
Initial	55	--
After weathering	45	--
After low temperature	50	--
Dynamic absorption, percent:		
Initial	--	25
After weathering	--	25

3.6.2 Flame resistance. The cloth shall be flame resistant treated. The finished cloth shall conform to the requirements specified in table V when tested as specified in 4.2.3 and shall show no evidence of melting or dropping of flaming pieces at any time during flame resistance testing.

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TABLE V. Flame resistance requirements

Characteristic	Requirement (maximum) Warp	Filling
After-flame time, seconds (Method 5903):		
Initial	2.0	2.0
After weathering	2.0	2.0
After-flame time, seconds (Method 5905):		
Initial	2.0	2.0
After weathering	2.0	2.0
Char length, inches (Method 5903):		
Initial	5.0	5.0
After weathering	5.0	5.0
Percent consumed (Method 5905):		
Initial	50	50
After weathering	50	50

3.6.3 Flexibility. The finished cloth shall show a bending moment of not more than 0.013 inch-pounds initially and not more than 0.032 inch-pounds after an exposure of minus 20° ± 5°F and after heating at 200° to 205°F, when tested as specified in 4.2.3.

3.7 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "face" at each end of the roll.

3.8 Fiber identification. Each roll shall be labeled and ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.9 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be in continuous pieces, each not less than 40 yards in length and shall be put-up in rolls in accordance with 5.1.

3.10 Workmanship. The finished cloth shall conform to the quality established by this document. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point values.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise

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specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Components and materials inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item examination.

4.2.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. when the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 with the following exceptions: glossary numbers 1, 2, 3, 5, 14, 26, 27, and 29 shall not be scored unless they exceed three times the thickness of the yarn. Misdraws and reed marks shall be scored if they result in a clearly noticeable separation of warp yarns. Only those slubs and knots which exceed three times the normal yarn size shall be scored. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll with the sample shall be penalized more than 4 points. The sample size shall be 20 rolls. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 40.0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceeds 60.0 points. If one roll exceeds 60.0 points per 100 square yards, a second sample of 20 rolls shall be examined only for individual roll quality. The lot shall be unacceptable if one or more rolls in the second sample exceeds 60.0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.2.2.1.1 Demerit points. Demerit points shall be assigned as follows:

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For defects 3 inches or less in any dimension	- one defect
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur:

Hole, cut or tear.
Objectionable odor.
Baggy, ridgy or wavy cloth.
Width less than specified or cut raw edge.
Edge ravels when pulled outward.
Slack or tight selvages 1/.
Overall uncleanness.
Poorly constructed - not firmly or tightly woven.
Overall application of compound not uniform.
Tackiness.

1/ To determine the presence of unacceptable selvage conditions, the following procedure shall be observed: During the visual examination, the perch shall be stopped a minimum of three times for each roll in the sample, the tension removed and the finished cloth examined for the selvage conditions. Suspect rolls should be removed from the perch and unrolled on the floor or a table and further examined for the condition. A waviness in the selvage or significant ripples diagonally across the width of the fabric is an indication of slack or tight selvages.

4.2.2.2 Length examination.

4.2.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for put-up and length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the roll ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the roll tickets. The rolls examined shall be those selected for examination of individual rolls.

4.2.2.2.3 Roll marking examination. During the yard-by-yard examination each roll in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

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Face stamping missing from either or both ends.
Face stamping on wrong side.
Not labeled, or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

4.2.3 End item testing. The cloth shall be tested for the characteristics listed in table VI. The methods of testing specified in FED-STD-191, wherever applicable and as listed in table VI shall be followed. The physical and chemical values specified in section 3 apply to the results of the determinations made on a sample shall be 15 continuous yards full width. The sample size shall be as shown below. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. Unless otherwise specified herein, all physical tests shall be performed under Standard Conditions as specified in FED-STD-191. Test reports shall contain the individual values utilized in expressing the final results.

Lot size (yards)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE VI. End item tests

Characteristics	Requirement paragraph	Test method
Material identification	3.2.1	1/
Weight	3.3	5041
Yarns per inch	3.3	5050
Breaking strength	3.3	5100
Tearing strength	3.3	ASTM D 1424
Air permeability	3.3	5450
Gloss (face side only)	3.3	ASTM D 523
Weave	3.3.1	Visual
Spectral reflectance	3.5.1.1	4.3.1
Colorfastness to:		
Weathering (after 300 hours)	3.5.2	5671
Crocking	3.5.2.1	5651 2/

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TABLE VI. End item tests (cont'd)

Characteristics	Requirement paragraph	Test method
Water repellancy:		
Spray rating:		
Initial	3.6.1	5526
After weathering	3.6.1	4.3.2 and 5526
Hydrostatic resistance:		
Initial	3.6.1	5514
After weathering	3.6.1	4.3.2 and 5514
After low temperature	3.6.1	4.3.3
Dynamic absorption:		
Initial	3.6.1	5500
After weathering	3.6.1	4.3.2 and 5500
Flame-resistance 3/:		
After-Flame time (Method 5903):		
Initial	3.6.2	5903
After weathering	3.6.2	4.3.4 and 5903
After-flame time (Method 5905):		
Initial	3.6.2	5905
After weathering	3.6.2	4.3.4 and 5905 4/
Char length:		
Initial	3.6.2	5903
After weathering	3.6.2	4.3.4 and 5903
Percent consumption:		
Initial	3.6.2	5905 4/
After weathering	3.6.2	4.3.4 and 5905 4/
Flexibility:		
Initial	3.6.3	4.3.5
At low temperature	3.6.3	4.3.6
After heat aging	3.6.3	4.3.7

1/ A certificate of compliance is required and will be acceptable for the requirement stated.

2/ AATCC Chromatic transference scale.

3/ For each flame-resistance test, ten specimens (five cut in warp direction and five in filling direction) shall be tested.

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4/ Except that only one 12 second flame immersion time shall be used.

4.2.4 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of PPP-P-1135.

4.3.1 Spectral reflectance test. Reflectance data shall be obtained from 600 to 860 nanometers (nm) for Forest Green 433 and from 700 to 860 nm for Desert Tan 459 relative to barium sulfate standard white reference or a set of reference tiles (see 6.6). The spectral band width at 860 nm shall be less than 25 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode operation. When the polychromatic mode operation is used, the spectrophotometer (see 6.7) shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. Specimens shall be measured as a single layer backed with two layers of the same shade cut from the standard. Readings will be taken on a minimum of two different areas, and the data averaged. The specimen shall be viewed at an angle no greater than 10 degrees from normal. Photometric accuracy of the spectrophotometer shall be within 1 percent and the wavelength accuracy shall be within 2 nm. When the measured reflectance values for any color at four or more of the listed wavelengths do not meet the limits specified in table II or III as applicable it shall be considered a test failure.

4.3.2 Accelerated weathering procedure for water resistance tests. The specimens for the water resistance after weathering tests shall be prepared and weathered in accordance with 4.3.4.

4.3.3 Hydrostatic resistance after low temperature test. The specimens shall be exposed to low temperature as specified in Method 5874 of FED-STD-191 except that the exposure temperature shall be $-20^{o} \pm 5^{o}F$ and the exposure time shall be a minimum of 4 hours. The face side on the specimens shall be toward the outside of the fold. After exposure to the low temperature, the specimens shall be tested for hydrostatic pressure resistance as specified in Method 5514 of FED-STD-191 with the water pressure applied to the face side of the specimens.

4.3.4 Accelerated weathering procedure for flame resistance tests. The apparatus and procedure shall be in accordance with Method 5804 except that the filters shall be removed and the weathering procedure shall be as follows. Two swatches, each 9-1/2 by 28 inches, shall be cut from the sample unit. One swatch shall be cut with the long dimension in the direction of the warp; the other with the long dimension in the direction of the filling. When the width of the sample unit does not permit a full 28-inch swatch to be taken in the filling direction, two 9-1/2 by 14-inch swatches shall be cut. The swatches shall be exposed one above the other, in the quadrant of the accelereated weathering apparatus for 100 hours. The swatches shall be changed from the top to bottom racks and visa versa each time the carbons are changed (approximately 17 to 20 hours) during a 100-hour exposure period. At the conclusion of the 100-hour exposure period, the swatches shall be removed from the apparatus and allowed to dry. Then five specimens for the fire resistance test shall be cut

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from each exposed swatch. All specimens shall be conditioned at Standard Conditions prior to testing. The five specimens shall be cut such that the direction being tested shall have been exposed to accelerated weathering in the vertical position. In conduction fire resistance tests on these specimens, three specimens shall be subjected to the flame at the end which was at the top of the exposed material, and two shall be subjected to the flame at the end which was at the bottom of the specimen. The lower edges of the fire resistance test specimens shall be trimmed, if necessary, so that a freshly cut end is exposed to the test flame.

4.3.5 Flexibility test. The flexibility test shall be in accordance with Method 5202 of FED-STD-191 except that eight specimens, four with the long dimension in the warp direction and four in the filling direction, shall be cut from the sample unit and pressed between two 6 by 6-inch glass plates weighted with a 20-pound weight for 4 hours at Standard Conditions prior to determining the flexibility. The load scale reading shall be taken only at a 20 degree angular deflection for each specimen.

4.3.6 Flexibility at low temperature test. Flexibility test specimens shall be prepared as specified in 4.3.5. The weight and upper glass plate shall then be removed. The lower plate holding the conditioned and pressed specimens and the instrument shall than be subjected to a temperature of minus 20° ± 5°F for not less than 1 hour and then tested, at that temperature, as specified in Method 5202 of FED-STD-191.

4.3.7 Flexibility after heat aging test. Flexibility test specimens shall be cut as specified in 4.3.5. The specimens shall be subjected to a temperature of 200° to 205°F for 120 hours in a well ventilated oven and then allowed to cool at Standard Conditions for 16 hours. The heat aged and conditioned specimens shall then be placed on glass plates under a 20-pound weight for 4 hours prior to testing as specified in Method 5202 of FED-STD-191.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1135.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1135.

5.3 Marking. In addition to any special marking required in the contact or purchase order, shipments shall be marked in accordance with the requirements of PPP-P-1135.

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6. NOTES

6.1 Intended use. Polyester duck, treated for fire, water, and weather resistance, is intended for use in manufacture of tents and tent flies.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Width of cloth required (see 3.4).
c. Color required (see 3.5).
d. Water resistance requirement, if other than specified (see 3.6.1).
e. Length, of other than specified (see 3.9).
f. Selection of applicable levels of put-up, preservation and packing (see 5.1 and 5.2).
g. Type of marking required (see 5.3).

6.3 Standard sample. For access to standard sample, address the contracting activity issuing the invitation for bids.

6.4 Formula approval. Approval of formulations is the responsibility of the U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 and is based on extensive tests including those for toxicity which are not set forth in this document. Because of the time required to evaluate and approve new treatments (approximately 6 months) only those chemical treatments already approved will be considered acceptable for the related procurement. Information pertaining to approval of new treatments should be obtained from the U.S. Army Natick Research and Development Center. The list of approved treatments may be obtained from the contracting activity.

6.5 Approval of flame resistants. As a part of the approval testing of new flame retardant finishes or fiber for tentage use, the finished cloth is required to have an after-flame time of not more than 2 seconds and not melt or drop flaming pieces when tested in accordance with Method 5905 of FED-STD-191, using one 12-second flame immersion.

6.6 Source of material. Barium sulphate of suitable quality for use as a white standard is available from Eastman Kodak Company. Suitable gray and white tiles are available from the National Bureau of Standards or Spectrophotomter manufacturers (see 4.3.1).

6.7 Spectrophotometers. The basic instrument for measuring the infrared reflectance is the General Electric Diano Hardy spectrophotometer, but other instruments such as the Beckham DU or DK, Diano Match-Scan, the Cary 14 or the Hunter D54P-IR spectrophotometers can be used (see 4.3.1).

6.8 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-44103A	Cloth, Duck, Polyester, Fire, Water and Weather Resistant

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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